

The smooth spherical-wavelet asymptotic, and a phase counterexample to the literal formula

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Abstract

Remark 7.4 of Dahlke–De Mari–De Vito–Hansen–Hasannasab–Quellmalz–Steidl–Teschke conjectures an asymptotic for a zonal spherical-wavelet coefficient and identifies the missing ingredient as an integrable bound after the rescaling $c = \ell a$. We supply that bound for every axisymmetric $\eta \in C_c^\infty((0, \pi); \mathbb{C})$ satisfying the paper’s cancellation condition. The resulting formula has an absolute square on the right. For real-valued profiles this proves the displayed conjecture on a natural strong regularity class. For complex-valued profiles the ordinary square printed in both the arXiv and published versions is false: multiplication by i leaves the left side invariant and reverses the sign of the right side.

Status and scope. The positive theorem below assumes axisymmetry, smoothness, and compact support away from the two poles. It does not claim the weakest possible hypotheses. The phase counterexample, however, is a complete disproof of the formula as written on the paper’s complex L_2 -space, even under these very strong regularity assumptions. Equivalently, the displayed conjecture is correct in this class for real profiles and becomes correct for complex profiles after replacing the final square by an absolute square.

1 Statement

Use the notation of the source paper: D_a is its spherical dilation, $\eta_a = D_a \eta$, Y_ℓ^0 is the normalized zonal spherical harmonic, and $\kappa_0(\theta) = \tan(\theta/2)$. For an axisymmetric profile, its zonal average $\tilde{\eta}$ is simply η .

Theorem 1 (Smooth corrected form of Remark 7.4). *Let $\eta \in C_c^\infty((0, \pi); \mathbb{C})$ be axisymmetric and satisfy*

$$\int_0^\pi \tan(\theta/2) \eta(\theta) d\theta = 0. \quad (1)$$

Then

$$\lim_{\ell \rightarrow \infty} \frac{1}{2\ell + 1} \int_0^\infty |\langle Y_\ell^0, \eta_a \rangle|^2 \frac{da}{a^3} = 4\pi \int_0^\infty \left| \int_0^\pi \tan(\theta/2) J_0(2c \tan(\theta/2)) \eta(\theta) d\theta \right|^2 \frac{dc}{c}. \quad (2)$$

The integral on the right is finite. It is strictly positive when $\eta \neq 0$.

For real η , the absolute square in (2) is the ordinary square printed in Remark 7.4, so this is an affirmative result for that conjecture under explicit stronger assumptions.

2 Idea of the proof

Put $t = \tan(\theta/2)$, so the support of η becomes a compact interval $[r, R] \Subset (0, \infty)$, and rescale $a = c/\ell$. The source already proves pointwise convergence of the inner integral to a Hankel transform. What is needed is tightness in the logarithmic measure dc/c .

There are four regimes. Cancellation makes the inner integral $O(c^2)$ near $c = 0$. Up to $c = \delta\ell$, the uniform Hilb formula turns the Legendre polynomial into an oscillatory Bessel function; integration by parts gives arbitrary inverse powers of c , apart from a harmless $O(\ell^{-3/2})$ remainder. When c/ℓ stays in a compact subinterval of $(0, \infty)$, the phase is nonstationary at frequency ℓ . Finally, for c/ℓ large, the elementary dilation amplitude is $O((\ell/c)^2)$. These estimates exactly close the gap stated by the authors.

3 Proof of the asymptotic

Proof of Theorem 1. Set

$$h(t) = \eta(2 \arctan t), \quad A(t, a) = \frac{2t}{(1+t^2)(1+a^2t^2)}, \quad z(t, a) = 2 \arctan(at).$$

The calculation in Steps 1–2 of the source proof of Theorem 7.3, with absolute values retained for complex functions, gives

$$\frac{1}{2\ell+1} \int_0^\infty |\langle Y_\ell^0, \eta_a \rangle|^2 \frac{da}{a^3} = 4\pi \int_0^\infty |I_\ell(a)|^2 \frac{da}{a}, \quad (3)$$

$$I_\ell(a) = \int_r^R A(t, a) P_\ell(\cos z(t, a)) h(t) dt. \quad (4)$$

Here h is extended by zero, so it and all its derivatives vanish at the chosen endpoints. Condition (1) becomes

$$I_\ell(0) = \int_r^R \frac{2t}{1+t^2} h(t) dt = 0. \quad (5)$$

After $a = c/\ell$, define

$$F(c) = \int_r^R \frac{2t}{1+t^2} J_0(2ct) h(t) dt.$$

The Mehler–Heine/Hilb limit used in the source yields

$$I_\ell(c/\ell) \longrightarrow F(c) \quad \text{for every fixed } c > 0. \quad (6)$$

We now prove the missing uniform tightness. First, the standard bounds are

$$\|P_\ell\|_{L_\infty[-1,1]} \leq 1, \quad \|P'_\ell\|_\infty \leq \ell^2,$$

the second being Markov's inequality. Since

$$1 - \cos z(t, a) = \frac{2a^2t^2}{1+a^2t^2}, \quad A(t, a) - A(t, 0) = O(a^2)$$

uniformly on $[r, R]$, equations (5)–(4) imply

$$|I_\ell(c/\ell)| \leq Cc^2, \quad 0 < c \leq 1. \quad (7)$$

Second, the uniform Hilb formula quoted in the source is

$$P_\ell(\cos z) = \left(\frac{z}{\sin z}\right)^{1/2} J_0((\ell + \tfrac{1}{2})z) + O_\varepsilon(\ell^{-3/2}), \quad 0 \leq z \leq \pi - \varepsilon. \quad (8)$$

For $q \geq 1$, write the standard Bessel expansion as

$$J_0(q) = e^{iq} b_+(q) + e^{-iq} b_-(q), \quad |b_\pm^{(k)}(q)| \leq C_k q^{-1/2-k}.$$

Choose $\delta > 0$ with $2 \arctan(\delta R) < \pi/2$. If $1 \leq c \leq \delta\ell$, then the phase

$$\Phi_{\ell,c}(t) = (\ell + \tfrac{1}{2})z(t, c/\ell)$$

satisfies $\Phi'_{\ell,c}(t) \asymp c$, uniformly on $[r, R]$; all normalized phase and amplitude derivatives are uniform as well. Repeated integration by parts in the two exponentials, followed by (8), gives, for every N ,

$$|I_\ell(c/\ell)| \leq C_N c^{-N} + C \ell^{-3/2}, \quad 1 \leq c \leq \delta \ell. \quad (9)$$

Fix $A > \delta$. When $\delta \ell \leq c \leq A \ell$, the parameter $a = c/\ell$ lies in $[\delta, A]$. Hence $z(t, a)$ stays in a compact subinterval of $(0, \pi)$, and

$$\partial_t z(t, a) = \frac{2a}{1 + a^2 t^2}$$

is uniformly separated from zero. One integration by parts in the Bessel waves in (8), including their $q^{-1/2}$ amplitude, gives

$$|I_\ell(c/\ell)| \leq C_A \ell^{-3/2}, \quad \delta \ell \leq c \leq A \ell. \quad (10)$$

For the remaining range, $a = c/\ell \geq A$, the support condition $t \geq r$, $|P_\ell| \leq 1$, and (4) give the elementary bound

$$|I_\ell(c/\ell)| \leq C a^{-2} = C(\ell/c)^2, \quad c \geq A \ell. \quad (11)$$

The limiting transform has the analogous endpoint bounds

$$|F(c)| \leq C c^2 \quad (0 < c \leq 1), \quad |F(c)| \leq C_N c^{-N} \quad (c \geq 1). \quad (12)$$

The first follows from cancellation and $J_0(2ct) - 1 = O(c^2 t^2)$; the second follows from the same Bessel-wave integration by parts, since h is smooth and compactly supported away from zero. Thus $|F(c)|^2/c$ is integrable.

For completeness, the estimates assemble as follows. For $M > 1$,

$$\begin{aligned} \limsup_{\ell \rightarrow \infty} \int_M^{\delta \ell} |I_\ell(c/\ell)|^2 \frac{dc}{c} &\leq C_N M^{-2N}, \\ \int_{\delta \ell}^{A \ell} |I_\ell(c/\ell)|^2 \frac{dc}{c} &\leq C_A \ell^{-3} \log(A/\delta) \rightarrow 0, \\ \int_{A \ell}^{\infty} |I_\ell(c/\ell)|^2 \frac{dc}{c} &\leq C A^{-4}. \end{aligned}$$

On $(0, M]$, equations (6) and (7) give dominated convergence (split once at $c = 1$). Let first $\ell \rightarrow \infty$, then $M \rightarrow \infty$, and finally $A \rightarrow \infty$. We obtain

$$\int_0^\infty |I_\ell(c/\ell)|^2 \frac{dc}{c} \rightarrow \int_0^\infty |F(c)|^2 \frac{dc}{c}.$$

Together with (3), this proves (2).

Finally, put $g(t) = 2h(t)/(1 + t^2)$. Then $F(c) = \int_0^\infty t g(t) J_0(2ct) dt$. Injectivity (or Parseval's theorem) for the order-zero Hankel transform shows that $F \not\equiv 0$ whenever $\eta \not\equiv 0$. Continuity of F then makes the right side of (2) strictly positive. $\square$

4 A full counterexample to the printed complex formula

Corollary 1. *There is a nonzero axisymmetric $\eta \in C_c^\infty((0, \pi); \mathbb{C})$ satisfying every explicit assumption of the source theorem and its cancellation condition, for which the equality printed in Remark 7.4 is false.*

Proof. Choose nonzero nonnegative $u, v \in C_c^\infty((0, \pi); \mathbb{R})$ with disjoint supports, and set

$$m_u = \int_0^\pi \tan(\theta/2) u(\theta) d\theta, \quad m_v = \int_0^\pi \tan(\theta/2) v(\theta) d\theta, \quad \eta_0 = u - \frac{m_u}{m_v} v.$$

Then $m_u, m_v > 0$, $\eta_0 \neq 0$, and (1) holds. By Theorem 1, its absolute-square Hankel energy is a finite number $R > 0$.

Take $\eta = i\eta_0$. Every term on the left of the printed conjecture is unchanged, since

$$|\langle Y_\ell^0, i(\eta_0)_a \rangle|^2 = |\langle Y_\ell^0, (\eta_0)_a \rangle|^2.$$

Its limit is therefore R . But the ordinary-square right side printed in the paper is

$$4\pi \int_0^\infty (iF_0(c))^2 \frac{dc}{c} = -R < 0.$$

Thus the two sides cannot agree. The obstruction is removed either by assuming η real-valued or by printing an absolute square. $\square$

5 Source evidence

The published-author PDF states the missing domination issue and displays the ordinary square in Remark 7.4 (PDF page 23):

$$J_0 \leq \left| \int_0^\infty tg(t)J_0(2ct) dt \right| \leq \int_0^\infty |tg(t)J_0(2ct)| dt,$$

which in turn yields

$$\int_0^B \left| \int_0^\infty tg(t)J_0(2ct) dt \right|^2 \frac{dc}{c} \geq \frac{1}{B^2} \int_0^B c \left| \int_0^\infty tg(t)J_0(2ct) dt \right|^2 dc > 0.$$

Ultimately we conclude that (44) does not vanish so that

$$\liminf_{\ell \rightarrow \infty} G_\ell \geq 4\pi \liminf_{\ell \rightarrow \infty} \int_0^\infty f_\ell(a) \frac{da}{a} > 0. \quad \square$$

Remark 7.4. *We conjecture that (under slightly stronger assumptions on η) the limit indicated below exists, and*

$$\lim_{\ell \rightarrow \infty} \frac{1}{2\ell + 1} \int_0^\infty |\langle Y_\ell^0, \eta_a \rangle|^2 \frac{da}{a^3} = 4\pi \int_0^\infty \left(\int_0^\pi \kappa_0(\theta) J_0\left(2c \tan \frac{\theta}{2}\right) \tilde{\eta}(\theta) d\theta \right)^2 \frac{dc}{c}.$$

A proof of such a result would again require to determine an upper bound for $f_\ell(\frac{c}{\ell})$, but integrable w.r.t. $\frac{dc}{c}$, which in turn would lead to an alternative (and stronger) proof for the upper frame bound.

References

- [1] S. Dahlke, F. De Mari, E. De Vito, M. Hansen, M. Hasannasab, M. Quellmalz, G. Steidl, and G. Teschke, *Continuous Wavelet Frames on the Sphere: The Group-Theoretic Approach Revisited*, Appl. Comput. Harmon. Anal. **56** (2022), 123–149; arXiv:2012.13460. See Theorem 7.3 and Remark 7.4.
- [2] G. Szegő, *Orthogonal Polynomials*, 4th ed., American Mathematical Society, 1975, Theorem 8.21.6 (the uniform Hilb formula used by the source).
- [3] P. MacAulay-Owen, *Parseval's theorem for Hankel transforms*, Proc. London Math. Soc. **45** (1939), 458–474.